EMBEDDING COUNT IS AT MOST THE DEGREE, WITH EQUALITY EXACTLY FOR SEPARABLE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This lemma is the count used by lemma-separable-degree (row 20 of this pool, delivered L48344) and by the tensor-product embedding count.

Editorial note. The label lemma-separable-equality names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-separably-generated-separable (delivered card L48336: the embedding count equals the degree iff every step of the generating tower is separable), lemma-separable-goes-up (separability propagates down towers)

Source reference. Original reference 09H2: sources/stacks-fields/pinned-original-source.tex, source lines 1404–1404. Prerequisite pointer invoked at 1616; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030X: sources/stacks-fields/pinned-original-source.tex, source lines 1560–1560. Prerequisite pointer invoked at 1611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let K/F be a finite extension of fields. Let $\overline{F}$ be an algebraic closure of F. Then we have*

$$|\operatorname{Mor}_F(K, \overline{F})| \leq [K : F]$$

with equality if and only if K is separable over F.

Proof. This is a corollary of Lemma 030X. Namely, since K/F is finite we can find finitely many elements $\alpha_1, \ldots, \alpha_n \in K$ generating K over F (for example we can choose the α_i to be a basis of K over F). If K/F is separable, then each α_i is separable over $F(\alpha_1, \ldots, \alpha_{i-1})$ by Lemma 09H2 and we get equality by Lemma 030X. On the other hand, if we have equality, then no matter how we choose $\alpha_1, \ldots, \alpha_n$ we get that α_1 is separable over F by Lemma 030X. Since we can start the sequence with an arbitrary element of K it follows that K is separable over F. □

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